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T - Test

A sample of 10 students scored the following marks in an exam: [72, 68, 75, 70, 74, 69, 71, 73, 70, 72] We want to test whether the average mark = 70 (μ0 = 70) at 5% significance level using python

```
import numpy as np
from scipy import stats
marks = np.array([72, 68, 75, 70, 74, 69, 71, 73, 70, 72])
mu_0 = 70
t_stat, p_value = stats.ttest_1samp(marks, mu_0)
print(f"T-statistic: {t_stat:.3f}")
print(f"P-value: {p_value:.4f}")
alpha = 0.05
if p_value < alpha:
    print("Reject Null Hypothesis → Mean is significantly different
from 70.")
else:
    print("Fail to Reject Null Hypothesis → No significant
difference.")
```

```
T-statistic: 1.993
P-value: 0.0774
Fail to Reject Null Hypothesis → No significant difference.
```

Z - Test

```
import numpy as np
from math import sqrt
from scipy.stats import norm
x_bar = 51.2
mu_0 = 50
sigma = 3
n = 36
z_stat = (x_bar - mu_0) / (sigma / sqrt(n))
p_value = 2 * (1 - norm.cdf(abs(z_stat)))
print(f"Z-statistic: {z_stat:.3f}")
print(f"P-value: {p_value:.4f}")
alpha = 0.05
if p_value < alpha:
    print("Reject Null Hypothesis → Mean is significantly different
from 50 g.")
else:
    print("Fail to Reject Null Hypothesis → No significant
difference.")
```

```
Z-statistic: 2.400
P-value: 0.0164
Reject Null Hypothesis → Mean is significantly different from 50 g.
```

Anova Test

Three fertilizers (A, B, C) were tested on crop yield (in kg). Is there a significant difference among fertilizers? (Use α = 0.05)

Fertilizer Yields

A 20, 22, 23

B 19, 20, 18

C 25, 27, 26

```python
import numpy as np
from scipy import stats
A=[20,22,23]
B=[19,20,18]
C=[25,27,26]
f_stat,p_value=stats.f_oneway(A,B,C)
print(f"F-statistic: {f_stat:.3f}")
print(f"P-value: {p_value:.4f}")
alpha=0.05
if p_value<alpha:
    print("Reject Null Hypothesis-Means are significantly different.")
else:
    print("Fail to Reject Null Hypo
```